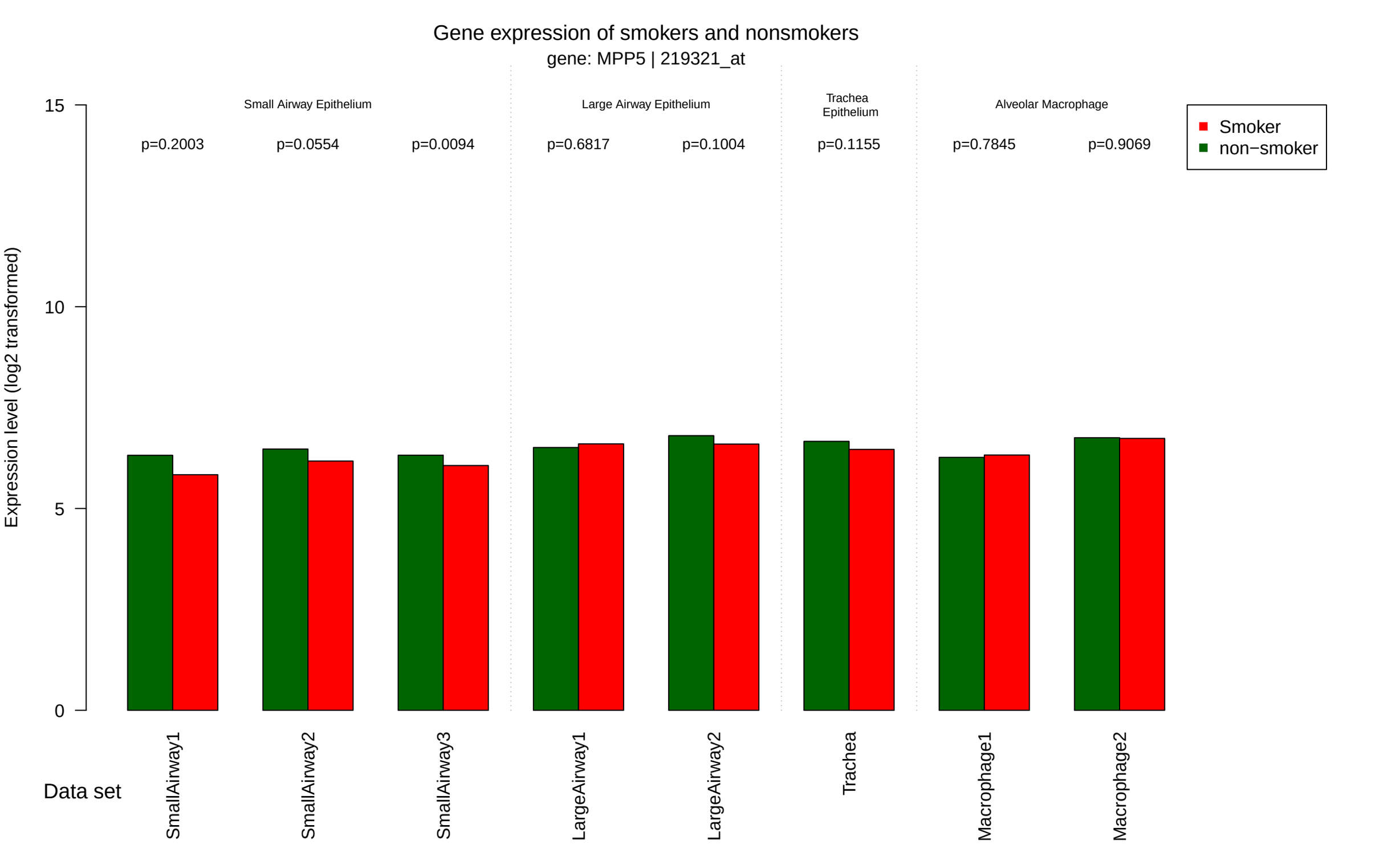
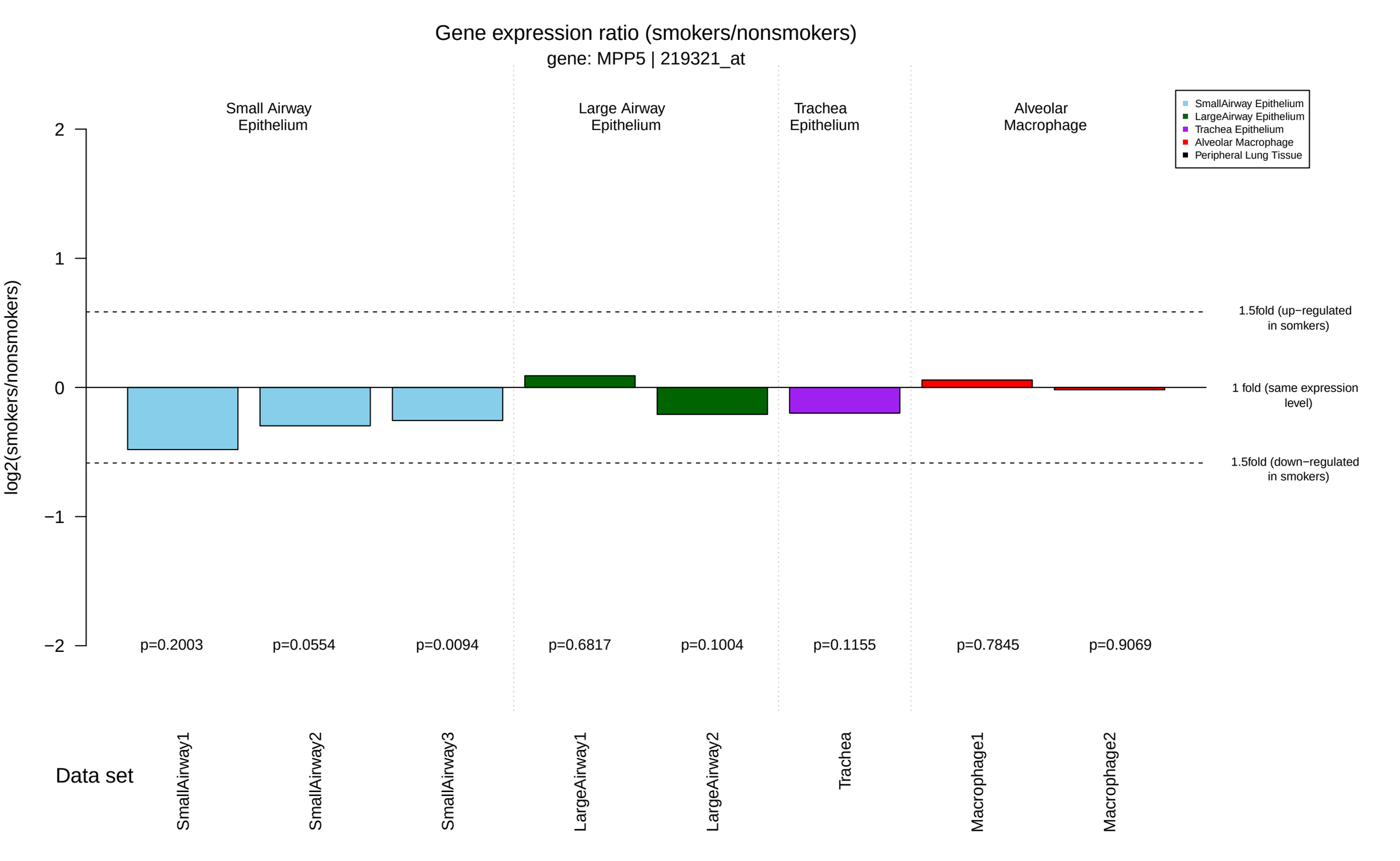
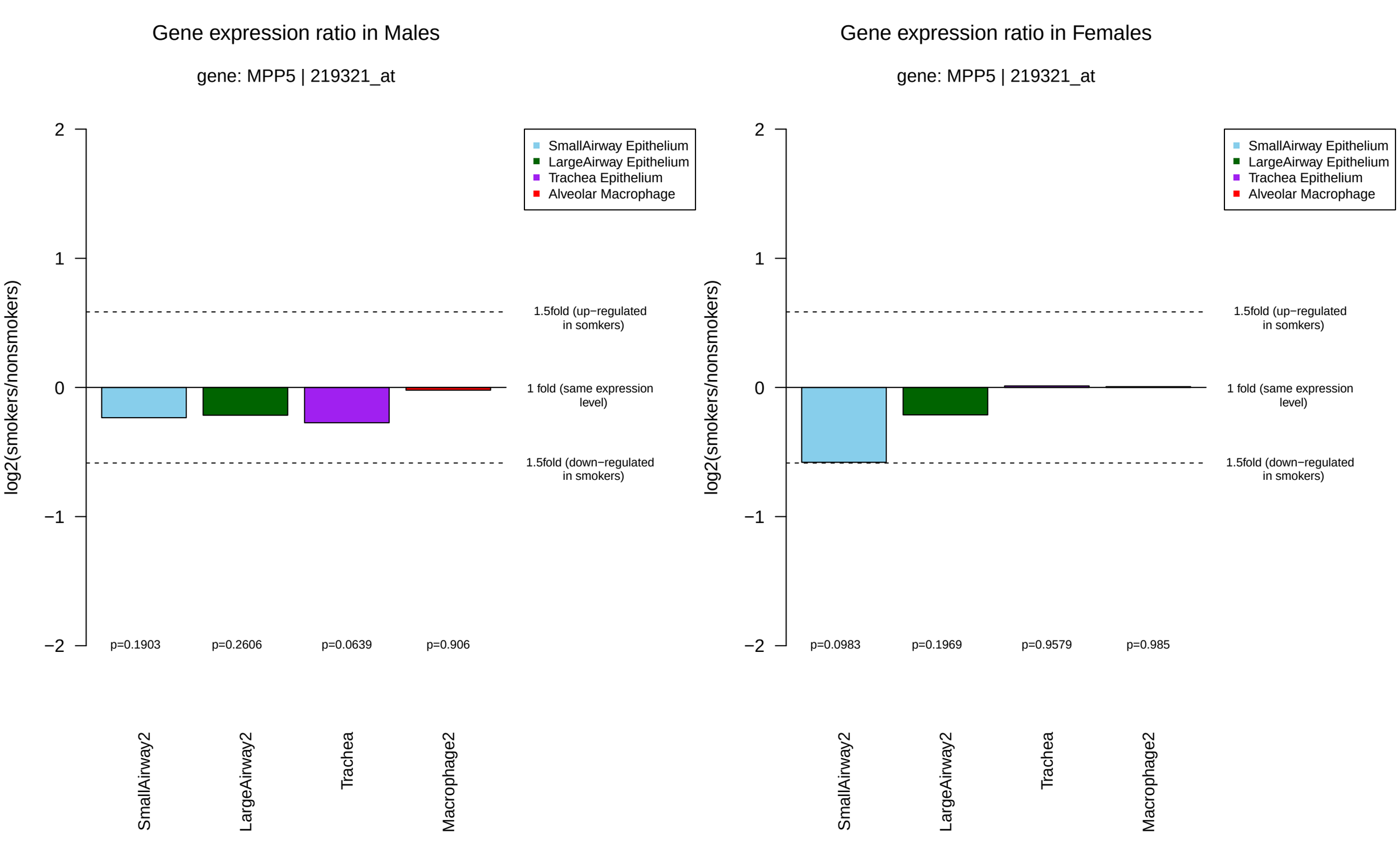
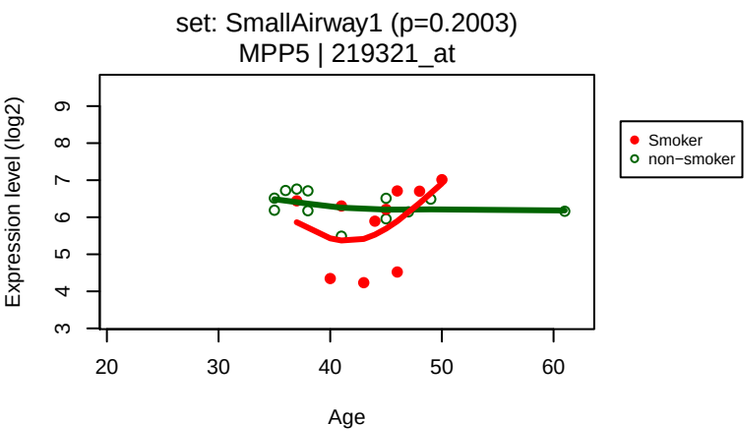
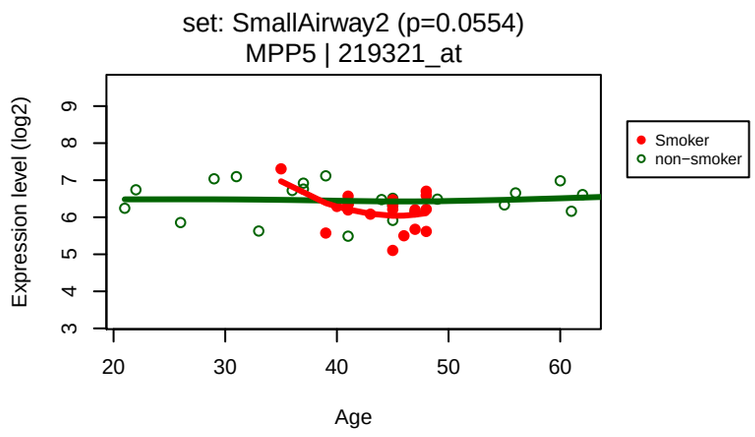
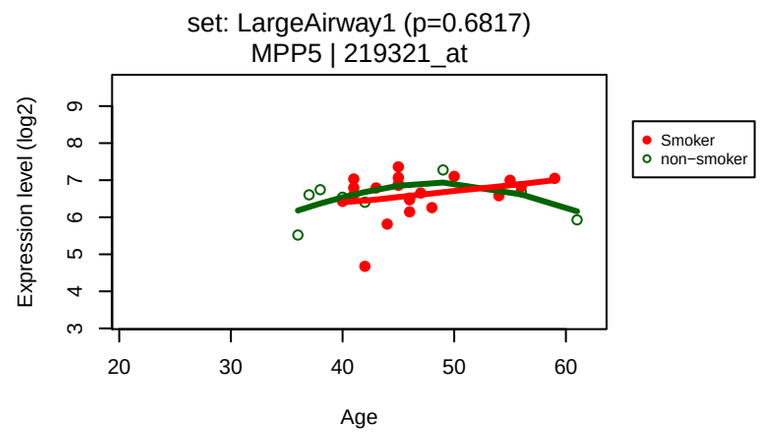

Gene expression of smokers and nonsmokers
gene: MPP5 | 219321_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2003
p=0.0554
p=0.0094
p=0.6817
p=0.1004
p=0.1155
p=0.7845
p=0.9069
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

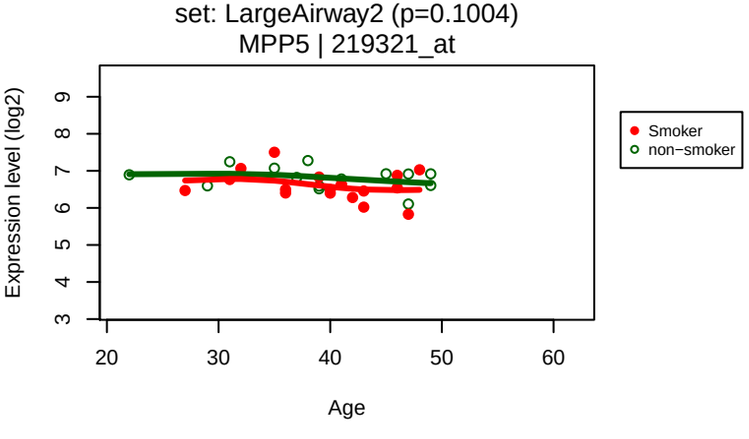

Gene expression ratio (smokers/nonsmokers)
gene: MPP5 | 219321_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2003
p=0.0554
p=0.0094
p=0.6817
p=0.1004
p=0.1155
p=0.7845
p=0.9069
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

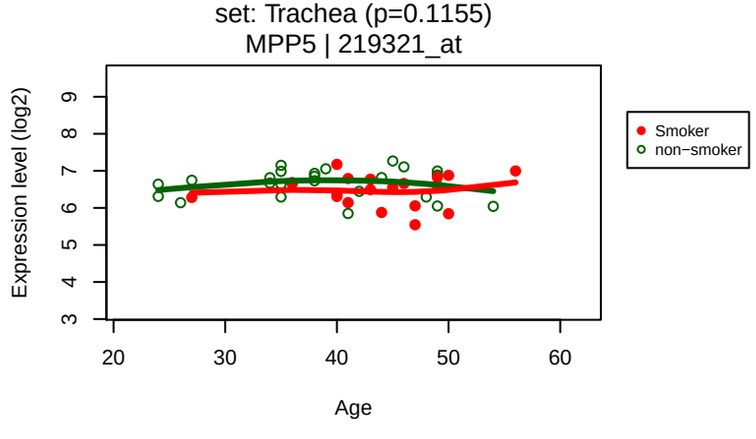

Gene expression ratio in Males
gene: MPP5 | 219321_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1903
p=0.2606
p=0.0639
p=0.906
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MPP5 | 219321_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0983
p=0.1969
p=0.9579
p=0.985
SmallAirway2
LargeAirway2
Trachea
Macrophage2

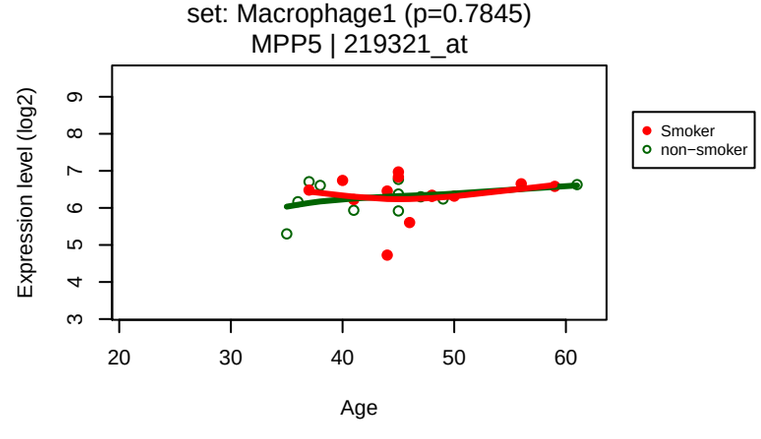

set: SmallAirway1 (p=0.2003)
MPP5 | 219321_at
Expression level (log2)
Age
Smoker
non−smoker

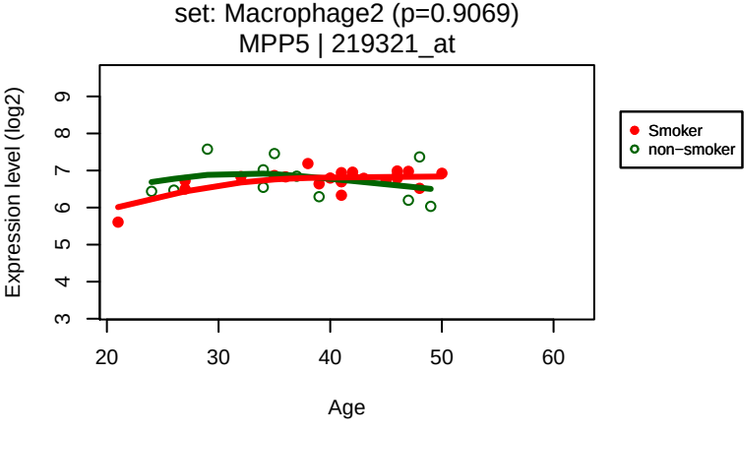

set: SmallAirway2 (p=0.0554)
MPP5 | 219321_at
Expression level (log2)
Age
Smoker
non−smoker

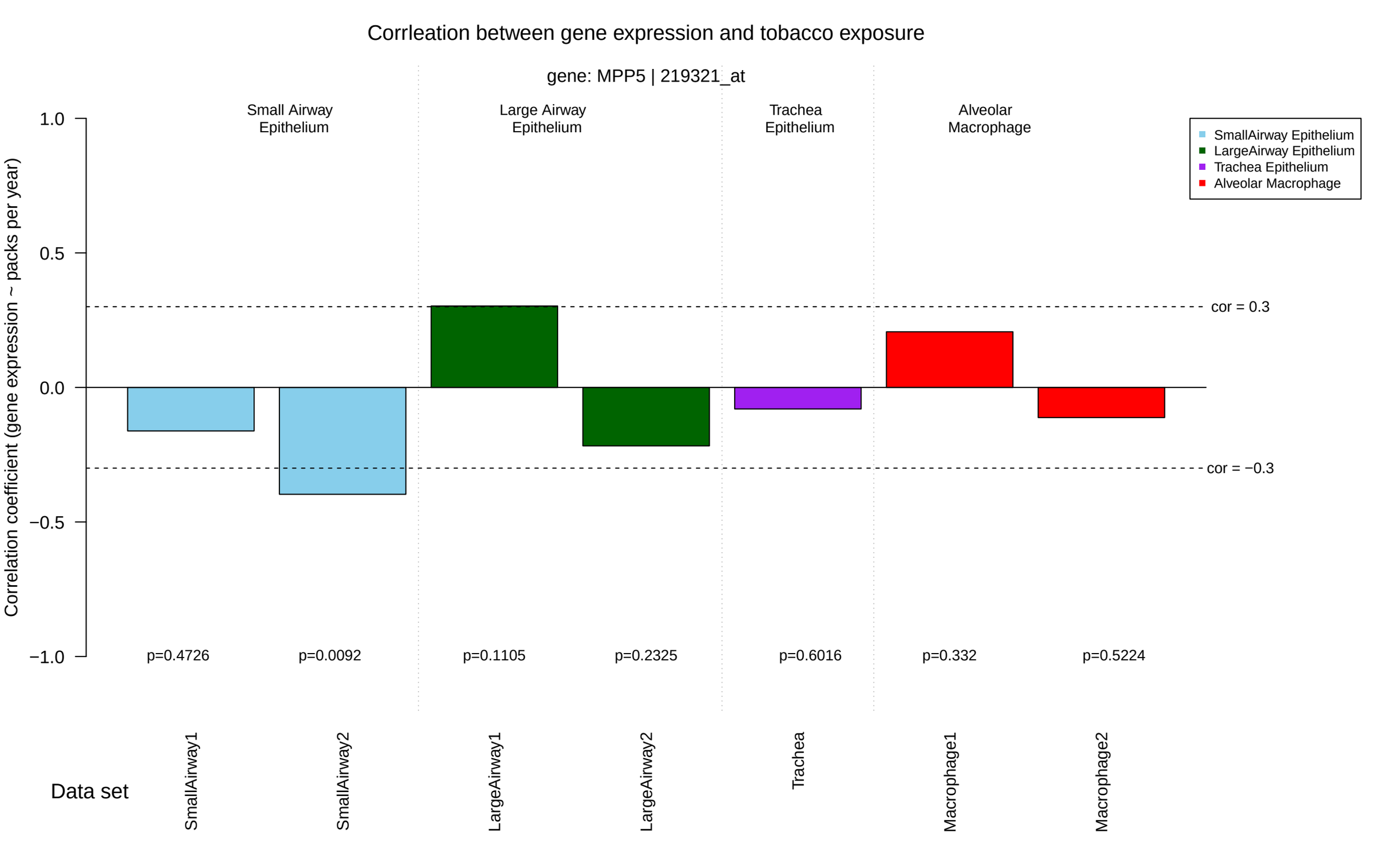

set: LargeAirway1 (p=0.6817)
MPP5 | 219321_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1004)
MPP5 | 219321_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1155)
MPP5 | 219321_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7845)
MPP5 | 219321_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9069)
MPP5 | 219321_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MPP5 | 219321_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4726
p=0.0092
p=0.1105
p=0.2325
p=0.6016
p=0.332
p=0.5224
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2